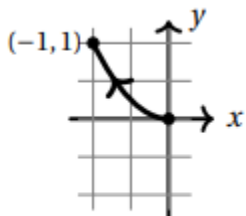
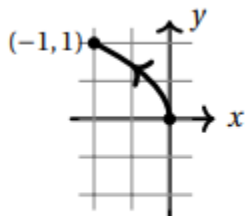
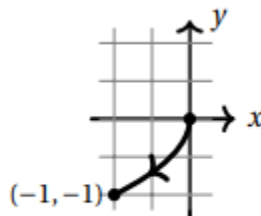
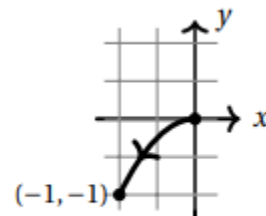


5.2.2 Line Integrals Worksheet

1. Let C be the curve in $\mathbb{R}^2$ parameterized by $\mathbf{r}(t) = \langle -t^2, t \rangle$ for $0 \leq t \leq 1$.

a) Mark the picture of C from among the choices below.


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b) For the vector field $\mathbf{F} = \langle y, -x \rangle$ compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

2. Find the work done by the force field $\mathbf{F}(x, y) = \langle y, x \rangle$ on a particle that moves along the cosine curve from $(0, 1)$ to $\left(\frac{\pi}{2}, 0\right)$.

3. Three continuous vector fields $\mathbf{F}, \mathbf{G}, \mathbf{H}$ on $\mathbb{R}^2$ are plotted below in the region with $-1 < x < 1$ and $-1 < y < 1$. Circle the best response for each of the following questions.

- a) Which vector field represents $\vec{\nabla} f$, where $f(x, y) = xy$.

F	G	H
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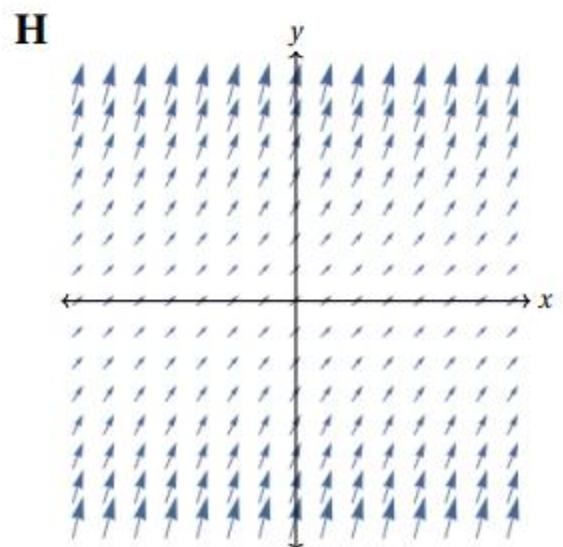
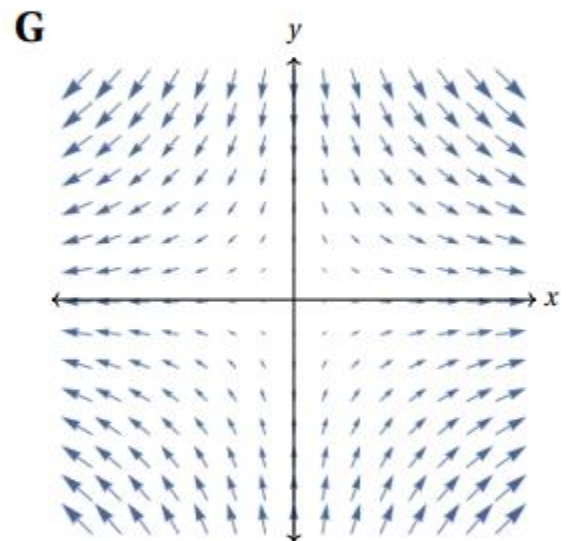
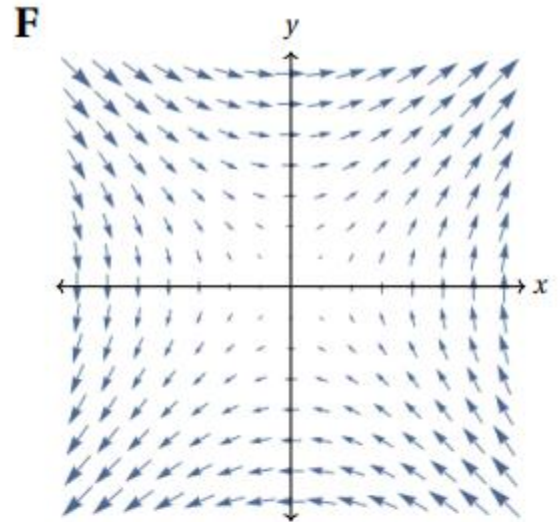
- b) A particle moves along a straight line from $(-1, -1)$ to $(1, 1)$. If the vector fields represent force fields, then:

The work done by $\mathbf{G}$ on the particle is...

positive	negative	zero
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The work done by $\mathbf{H}$ on the particle is...

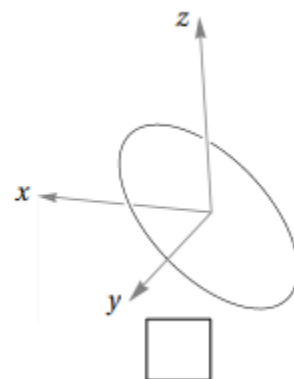
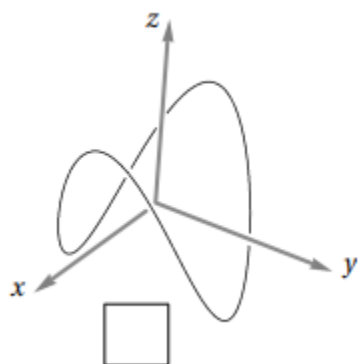
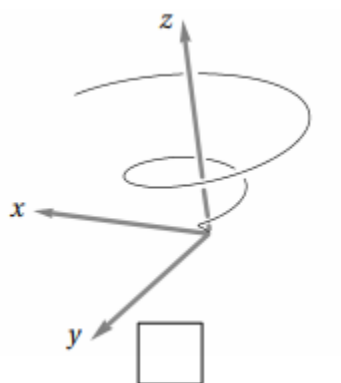
positive	negative	zero
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4. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ and C is the path along the circle of radius 3 starting at the point $(3, 0)$ and ending at the point $(0, -3)$ while moving clockwise.

5. Consider the curve C parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), \cos(2t) \rangle$ where $0 \leq t \leq 2\pi$.

- a) Check the box next to the correct sketch of C .



- b) Find the work done if a particle travels along path $\mathbf{r}(t)$ under the force given by the vector field $\mathbf{F}(x, y, z) = \langle -2y, 2x, 0 \rangle$.

6. Let C be the helix parameterized by $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ where $0 \leq t \leq 2\pi$.

a) Let $\mathbf{F}(x, y, z)$ be the vector field defined by $\mathbf{F}(x, y, z) = \langle x, y, 0 \rangle$. Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is oriented in the direction from $(1, 0, 0)$ to $(1, 0, 2\pi)$.

b) Now let $\mathbf{F}(x, y, z) = \langle ye^{x^2+y^2}, \ln(x^2 + y^2) - x, 1 \rangle$. Determine $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is given the opposite orientation.

7. Find the work done by the force field $\mathbf{F}(x, y) = \langle x^2, ye^x \rangle$ on a particle that moves along the parabola $x = y^2 + 1$ from $(1, 0)$ to $(2, 1)$ and then along the line segment from $(2, 1)$ to $(0, 0)$.

8. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \langle x - y, xy \rangle$ and C is the arc of the circle $x^2 + y^2 = 4$ traversed counterclockwise from $(2, 0)$ to $(0, -2)$.

- a) Evaluate this line integral in its current form using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.

b) Rewrite the integral in the form of $\int_C ? dx + ? dy$.

c) Evaluate the integral in the form of $\int_C ? dx + ? dy$.

9. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y, z) = \langle \sin(x), \cos(y), xz \rangle$ and C is given by the vector function $\mathbf{r}(t) = t^3\mathbf{i} - t^2\mathbf{j} + t\mathbf{k}$ where $0 \leq t \leq 1$.

a) Evaluate this line integral in its current form using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.

b) Rewrite the integral in the form of $\int_C ? dx + ? dy + ? dz$.

c) Evaluate the integral in the form of $\int_C ? dx + ? dy + ? dz$.

10. Consider the line integral $\int_C xy dx + 3y^2 dy$ where C is given by the vector function $\mathbf{r}(t) = \langle 11t^4, t^3 \rangle$ where

$$0 \leq t \leq 1.$$

a) Evaluate this line integral in its current form.

b) Rewrite the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$. What is $\mathbf{F}$? What is $d\mathbf{r}$?

c) Evaluate the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$ using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.

11. Consider the integral $\int_C (x+y)dx + (y-z)dy + z^2dz$ where C is given by the vector function

$$\mathbf{r}(t) = \langle t^2, t^3, t^2 \rangle \text{ where } 0 \leq t \leq 1.$$

a) Evaluate this line integral in its current form.

b) Rewrite the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$. What is $\mathbf{F}$? What is $d\mathbf{r}$?

c) Evaluate the integral in the form of $\int_C \mathbf{F} \cdot d\mathbf{r}$ using the formula $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt$.